

EXP.NO: 6

SharedMemory IPC

DATE:20/02/2026

AIM:

To implement Inter-Process Communication (IPC) using Shared Memory in Linux, where one process writes data to a shared memory segment and another process reads from it.

ALGORITHM:

1. Start the experiment.
2. Include required headers: `sys/ipc.h` and `sys/shm.h`.
3. Writer Process: Generate a unique key using `ftok()`. Create a shared memory segment using `shmget()`. Attach to the segment using `shmat()`. Write data into the shared memory buffer. Detach using `shmdt()`.
4. Reader Process: Generate the same key using `ftok()`. Access the existing segment using `shmget()`. Attach with `shmat()` and read the data. Detach using `shmdt()` and remove the segment using `shmctl(IPC_RMID)`.
5. Compile both programs and execute writer first, then reader.
6. Stop the experiment.

CODE:[illegible]

```

#include <stdlib.h>
#include <sys/ipc.h>
#include <sys/shm.h>

int main() {
    key_t key = ftok("shmfile", 65);
    int shmid = shmget(key, 1024, 0666 | IPC_CREAT);

    char *str = (char*) shmat(shmid, NULL, 0);

    printf("Data read from shared memory: %s", str);

    shmdt(str);
    /* Remove shared memory segment */
    shmctl(shmid, IPC_RMID, NULL);
    return 0;
}

```

OUTPUT:

```

# Terminal 1 - Writer
[hp@localhost ~]$ touch shmfile
[hp@localhost ~]$ gcc writer.c -o writer
[hp@localhost ~]$ ./writer
Write data: Hello from Shared Memory!
Data written to shared memory.

# Terminal 2 - Reader
[hp@localhost ~]$ gcc reader.c -o reader
[hp@localhost ~]$ ./reader
Data read from shared memory: Hello from Shared Memory!

```

RESULT:

Thus, Inter-Process Communication using Shared Memory was successfully implemented. The writer process wrote data into the shared memory segment and the reader process successfully read and displayed the data from the same segment.